

# Segmentation for Gaussian Graphical Models on Wastewater Sensor Data

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**Abstract.** The introduction of continuous monitoring sensors in wastewater treatment plants (WWTPs) offers an opportunity for fine-grained understanding and control. However, WWTP data is highly non-linear and non-stationary, with distinct regimes of operation. To address these issues we segment sensor data by regime. To infer causal relationships among sensors we perform graph decomposition, giving a temporal and a contemporaneous graph for each regime. To analyse the uncertainty in these graphs we evaluate the consistency within regimes and differences between regimes. They can then be used to communicate sensor relationships to domain experts and confirm expected relationships.

**Keywords:** Time Series · Multivariate · Segmentation · Granger Causality · Graphical Lasso · Graph Inference

## 1 Introduction

Wastewater treatment plants (WWTPs) take wastewater from homes, factories etc., and process it before returning it to water bodies. They constitute a vital piece of infrastructure in towns and cities. WWTPs sometimes fail to meet emission standards [10, 24, 8], leading to public health and environmental concerns [5, 19]. Improved monitoring and control contribute to environmental goals through reduced emissions and electricity use.

Such improvements are possible due to the prevalence of continuous monitoring sensors. This allows for key parameters to be monitored at a much higher granularity than was previously possible, but this abundance of data comes with challenges. Sensors require cleaning, maintenance, and calibration to operate correctly. They can be muddled, impacted by solids in the tank, or damaged, leading to spurious or missing data. WWTP processes can be highly non-linear and non-stationary, with periodicity across multiple time-scales and long-term trends. Importantly, the plant moves between several regimes of operation.

All of this makes for a process that is difficult to model, be it for understanding, forecasting or control. In this paper we attempt to address this difficulty by segmenting the time series according to regime of operation, giving segments which are more easily modelled. We seek a segmentation which reflects real

changes in the tank, and gives segments which are well-behaved according to some feature, e.g. provably stationary.

In this paper we propose two rules to segment the data. Both are control-chart methods, one using the running mean, and one using the running turning rate. To identify and communicate key sensor relationships within each regime, we propose to use a Gaussian Graphical Model (GGM), representing each regime by two graphs: a temporal one which captures autoregressive relationships, and an instantaneous one which captures the conditional dependence between sensors. We evaluate the segmentation and the graphical methods by studying consistency of edge weight distributions within regimes, and differences in distributions between regimes.

Section 2 reviews related work; Section 3 introduces the methodology. Sections 4-5 cover experiments and results. Section 6 discusses results and Section 7 the conclusion.

## 2 Background and Related Work

**Wastewater Treatment** The primary driver of emissions and energy consumption in WWTPs is the biological reactor. In this part of the plant biological nutrient removal (BNR) takes place. Excess nutrients such as nitrogen and phosphorus are removed from the water. This is done by managing bacteria populations and oxygen levels in the tank to select for strains of bacteria that metabolize or accumulate these pollutants. In this paper we focus on plant configurations that perform all BNR processes in one tank, rather than a series [6].

BNR processes are periodic across multiple time scales due to aeration control, daily patterns, and long term fluctuations in bacteria composition, temperature, etc. [14]. There are several main regimes of operation, distinguished by the amount of dissolved oxygen (DO) present, which is affected by whether aerators are switched on or off, and which strongly affects bacterial processes.

**Modelling Challenges.** Haimi et al. [13] note in their review of soft-sensor use in WWTPs that the non-linearity of process data means traditional multivariate statistical methods can be unsatisfying. Similarly Newhart et al. [18] remark that wastewater processes “exhibit nonlinear, nonstationary, autocorrelated, and co-correlated behavior” which makes data-driven modelling challenging. As the problem of segmentation is closely related to changepoint detection we also note the reviews of Li et al. and Xu et al. [16, 26]. Both authors note that evaluation is challenging with a distinct lack of ground truth labels. Simulations of BNR are available, e.g. BSM2 [11], but require substantial domain knowledge and parameter tuning.

**Time Series Segmentation Approaches.** There is substantial work in the field of Time Series Segmentation, with many mature and well-developed techniques available. PELT is a multi-changepoint detection method introduced by

Killick et al. [15] which seeks to minimize a cost function to detect multiple changepoints across a time series. This method can be constrained by poor parameter or cost function choices in applied settings. And segmentations based on statistical changes may not always coincide with true changes in BNR processes. Bayesian Online Changepoint Detection (BOCPD) is a changepoint detection method which encodes the time since a changepoint has occurred as runtime and updates a posterior distribution of this run length being zero [1]. This method has been applied in water quality monitoring [3] to detect faults in real time. Ba & McKenna apply an AR model to time series data and run online-changepoint detection on the residuals to detect contamination events. This approach is very similar to our own, but we wish to describe the multivariate structure within regimes too, rather than detect events in a binary fashion. There is substantial work in the related areas of clustering, with Assef et al. [2] writing a comprehensive review of techniques across the waste management sector. They mention that most clustering attempts rely on generic objective functions and methodologies. More case-specific applications may be more likely to be adopted by practitioners. We address this with rule-based regime detection based on WWTP operational knowledge, and graphical methods to provide an interpretable and engaging description of each regime. Rule-based regime detection has a long history in the field of wastewater management. For example, in 1993 Wareham et al. [25] proposed using Oxidation-Reduction Potential (ORP) to control anoxic sequences. There is little work integrating or using this domain knowledge to assist in data-driven modelling.

**Graphical Models for Time Series** Inferring graph structures from multivariate time series data has been done in several ways in the literature. In 1969 Granger proposed to measure the causal relationship between two variables in terms of predictive power [12]. Vector Autoregression (VAR) allows inference of Granger causality among multiple variables. The result can be represented as a graph where an edge  $X \rightarrow Y$  exists if the coefficient for  $X$  is meaningfully non-zero in prediction of  $Y$  in the VAR model [17]. Granger causality is typically based on linear prediction, but non-linear neural networks have also been used [21], which may be appropriate in the WWTP context. Granger causality belongs to the wider study of causal inference, using predictive power as evidence [23], with contemporary methods applied to time series data in [20].

An alternative method of graph inference is Graphical LASSO (GLASSO). Here a sparse *precision matrix* (inverse covariance matrix) is fitted to the data, using a penalization term to enforce sparsity. The precision matrix is a powerful tool as each entry contains the conditional dependence between variables: in essence, how much one variable influences the other, when conditioned on all others. This removes spurious edges caused by indirect effects. [9].

Epskamp et al. [7] proposed the Gaussian Graphical Model (GGM). Here Granger causality and Graphical LASSO's conditional dependence are described in the same model. The principle here is that after an effective VAR model is fit the residuals carry instantaneous relationships not explained by VAR. GLASSO

can be applied to the covariance matrix of these residuals to estimate the conditional dependencies remaining in the sensor data. The result is a temporal graph describing lagged relations, and a contemporaneous, or instantaneous graph describing the dependence structure of the sensors. Both graphs are estimated jointly using Multivariate Regression with Covariance Estimation (MRCE) [22]. The authors apply this to describe temporal relationships within and between subjects in psychological experiments. We will use GGMs to estimate WWTP sensor relationships per regime.

### 3 Methodology

Let  $x_t \in \mathbb{R}^q$  be the sensor data at time instance  $t \in [1, T]$  for  $q$  sensors. We propose to segment this data using two different methods, which shall be denoted by  $R_{\text{mean},3}$  and  $R_{\text{rank},5}$  henceforth. Both methods are based on the dissolved oxygen (DO) sensor, and produce 3 and 5 segment types respectively. Additionally we perform random segmentations  $R_{\text{rand},3}$ , and  $R_{\text{rand},5}$  as baselines for comparison.

A key difficulty is that many of the resulting segments will be too short for meaningful modelling. We address this by pooling the data across segments of a particular regime, using a block-matrix design for matrix VAR to avoid regression across segment boundaries, and bootstrapping to produce a distribution of graphs.

The assumption we wish to test is that the sensor data is consistent within a regime, and meaningfully distinct when comparing different regimes. To formalize this assumption we must describe how we pool our data, and the underlying model we use. For a particular regime having  $K$  distinct segments let  $x_t^{(k)} \in \mathbb{R}^q$  be the sensor reading at time  $t \in [1, T_k]$  in segment  $k \in [1, K]$ . Let

$$\mathbf{Y}^{(k)} = \begin{bmatrix} x_{p+1}^{(k)\top} \\ \vdots \\ x_{T_k}^{(k)\top} \end{bmatrix} \in \mathbb{R}^{(T_k-p) \times q}, \quad \mathbf{X}^{(k)} = \begin{bmatrix} x_p^{(k)\top} & \dots & x_1^{(k)\top} \\ \vdots & \ddots & \vdots \\ x_{T_k}^{(k)\top} & \dots & x_{T_k-p}^{(k)\top} \end{bmatrix} \in \mathbb{R}^{(T_k-p) \times qp} \quad (1)$$

where  $p$  is the maximum lag being considered. The first  $p$  sensor readings are dropped as target values to prevent incomplete regression, and also to prevent the regression from looking into non-contiguous segments. For a bootstrap sample of  $I \subset [1, K]$  regime instances we concatenate  $\mathbf{Y}^{(k)}$  and  $\mathbf{X}^{(k)}$  in the following way:

$$\mathbf{Y} = \begin{bmatrix} \mathbf{Y}^{k_1} \\ \vdots \\ \mathbf{Y}^{k_{|I|}} \end{bmatrix} \in \mathbb{R}^{N \times q}, \quad \mathbf{X} = \begin{bmatrix} \mathbf{X}^{k_1} \\ \vdots \\ \mathbf{X}^{k_{|I|}} \end{bmatrix} \in \mathbb{R}^{N \times qp} \quad (2)$$

where  $N = \sum_{k \in I} (n_k - p)$ . This can be used to perform VAR as follows:

$$\mathbf{Y} = \mathbf{X}\mathbf{B} + \mathbf{E} \quad (3)$$

which produces a matrix  $\mathbf{B} \in \mathbb{R}^{qp \times q}$  of coefficients, composed of  $p$  block matrices  $[\mathbf{B}^{(l)}]_{l \in [1,p]}$ , and a residual matrix  $\mathbf{E} \in \mathbb{R}^{N \times q}$  with inverse covariance matrix  $\Omega$ .

Matrices  $\mathbf{B}$  and  $\Omega$  give two graphs: first, a temporal graph which captures how sensors depend on each other within a window of time  $p$ , and second, an instantaneous graph which describes the remaining conditional dependence between sensors. These two matrices will be estimated using MRCE under sparsity constraints; details are provided in Section 3.2.

### 3.1 Segmentation

We propose two segmentation approaches. In the first we segment into sustained low-DO, sustained high-DO, and very low-DO (Denitrification) periods where bacterial denitrification is expected. In the second we distinguish between aeration being on or off. This gives three and five distinct regimes respectively. Note that all sensor data is sampled at 1-minute granularity.

**Mean-Based 3-way Segmentation** The mean-based segmentation rule, denoted by  $R_{\text{mean},3}$ , partitions the time series into three regimes using the local magnitude of the dissolved oxygen (DO) signal:

$$\text{Labels} = \{\text{low-DO}, \text{high-DO}, \text{denitrification}\}. \quad (4)$$

Let  $d_t$  denote the DO concentration at time  $t$ . To reduce short-term fluctuations, we use a centred rolling mean over a 60-minute window  $W_t$ :

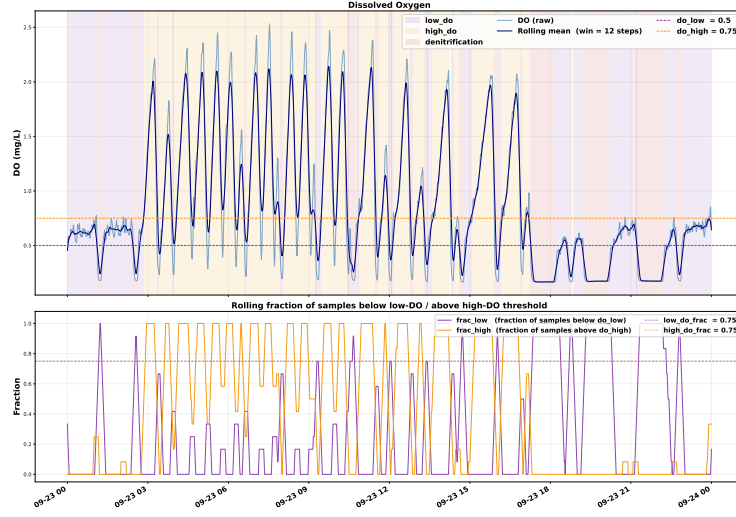
$$\bar{d}_t^{(w)} = \frac{1}{|W_t|} \sum_{i \in W_t} d_i. \quad (5)$$

Here,  $W_t$  is the set of time indices in the rolling window around  $t$ , and  $d_i$  is the DO value at index  $i$ . A point is assigned to low-DO when the rolling mean  $\bar{d}_t^{(w)}$  is below 0.5 mg/L and at least 75% of nearby observations also satisfy this low-DO condition. Similarly, a point is assigned to high-DO when the rolling mean is at least 0.75 mg/L and at least 75% of nearby observations satisfy the high-DO condition. This avoids assigning segment labels based on isolated spikes or short drops. Short unstable fragments are merged into the preceding segment using a minimum run length of 30 minutes.

The denitrification regime is handled with an override. Let  $\bar{d}_t^{(d)}$  be the rolling DO mean over a denitrification detection window. The denitrification label is applied when the DO signal remains below 0.30 mg/L for at least 20 minutes. The resulting segment is expanded by 15 minutes on both sides to include neighbouring transition points. This helps capture the boundary behaviour rather than only the most extreme low-DO points.

After these assignments, any remaining ambiguous regions are filled using neighbouring segment labels. The resulting segmentation gives a three-segment representation based on sustained DO conditions.

Fig. 1 shows a segmentation of a small subset of the data, using the  $R_{\text{mean},3}$  method.

Fig. 1: Mean-based three-way segmentation,  $R_{\text{mean},3}$ .

**Rank Based 5-way Segmentation** This segmentation method, denoted by  $R_{\text{rank},5}$ , distinguishes 3 long-term (in terms of hours) regimes, and for low-DO and high-DO regimes, distinguishes whether aeration is on or off, giving 5 regimes:

$$\text{Labels} = \{\text{high-o2-on}, \text{high-o2-off}, \text{low-o2-on}, \text{low-o2-off}, \text{anoxic-off}\} \quad (6)$$

In this setting high and low DO are identified in an ordinal or rank-based way, using zero crossing rate (ZCR), i.e. turning-rate. The ZCR is a powerful rank-based statistical tool that is robust to changes in range. For a formal review of turning-rate tests see Betken et al. [4].

Let  $x'_t$  be the differenced sensor series, and the ZCR be:

$$\text{ZCR}(x'_t) = \sum_{i=0}^{19} \mathbf{1}\{x'_{t-i} \cdot x'_{t-i-1} < 0\} \quad (7)$$

Define the functions to assign the high-DO and low-DO regimes as:

$$f^{\text{low}}(x_t) = \mathbf{1}\{\text{ZCR}(x'_t) \geq 7\}, \quad f^{\text{high}}(x_t) = \mathbf{1}\{\text{ZCR}(x'_t) < 7\} \quad (8)$$

i.e. if more than 7 out of the previous 20 differenced points cross the x-axis we say we are in the "high" regime. We denote anoxic as

$$f^{\text{anoxic}}(x_t) = \prod_{i=t-39}^t \mathbf{1}\{x_i < 0.15\} \quad (9)$$

as 0.15 is the largest DO sensor reading that can be considered anoxic, and these periods are programmed beforehand for a minimum of 40 minutes. We further

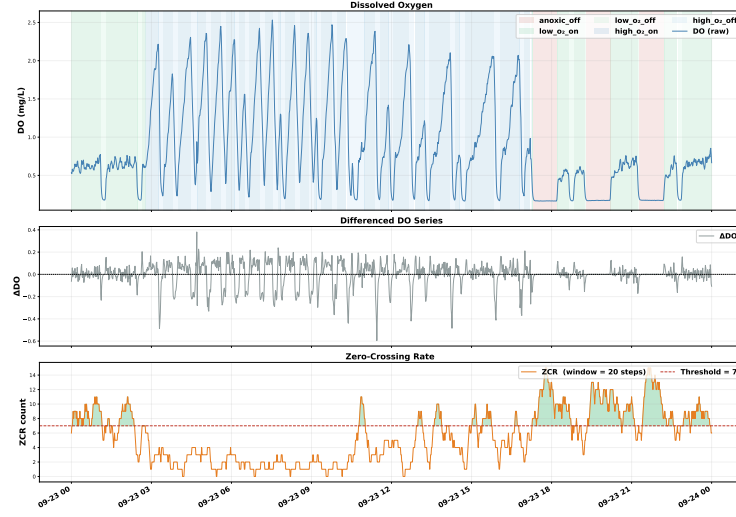


Fig. 2: Rank-based five-way segmentation,  $R_{\text{rank},5}$ , ZCR signal shown below.

differentiate these regimes by whether or not aeration is on. Since it is strictly off in the anoxic period this creates 5 regimes. The rules for aeration on and off are different for high-DO and low-DO:

$$f_{\text{on}}^{\text{high}}(x'_t) = \prod_{i=t-4}^t \mathbf{1}\{x'_i > 0\}, \quad f_{\text{off}}^{\text{high}}(x'_t) = \prod_{i=t-4}^t \mathbf{1}\{x'_i < 0\} \quad (10)$$

$$f_{\text{on}}^{\text{low}}(x_t) = \mathbf{1}\{x_t \geq 0.4\}, \quad f_{\text{off}}^{\text{low}}(x_t) = \mathbf{1}\{x_t < 0.4\} \quad (11)$$

We choose  $[t-4, t]$  as the blowers are switched on for a minimum of 10 minutes to reduce stress on the machinery. 5 consecutive increasing data points are the smallest amount that will reliably detect on/off cycles while tolerating a little noise. 0.4 is selected somewhat arbitrarily for reliable results. Fig. 2 shows results on the same data subset as before, using the  $R_{\text{rank},5}$  method.

**Random segmentation with K-Means Clustering** Random segmentation with K-means clustering is used as an unsupervised baseline. Segment boundaries are sampled uniformly from the data. Let  $S_j$  denote the set of data contained in the  $j$ th sampled segment. For each segment, a feature vector  $z_j$  is computed from the sensor variables in that segment. Composing  $z_j$  is  $\mu_{j,r}$  and  $\sigma_{j,r}$ , the mean and standard deviation of sensor  $r$  within window  $S_j$ . The variables are normalized to prevent larger numerical scales from dominating the clustering. These feature vectors  $z_j$  are then clustered using K-means.

We consider two versions of this segmentation, denoted by  $R_{\text{rand},3}$  and  $R_{\text{rand},5}$ , where the subscript gives the number of K-means clusters for comparison with our 3-way and 5-way segmentations. The cluster labels are not assigned a biological interpretation.

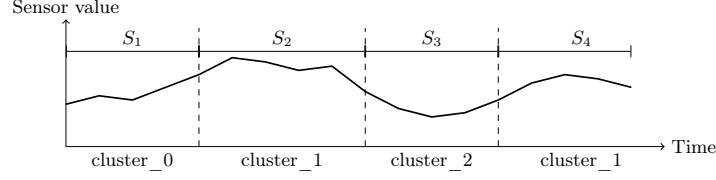


Fig. 3: Segments in the K-means clustering baseline. Each segment  $S_j$  is represented by a feature vector and assigned a K-means cluster.

### 3.2 Graph Decomposition

**MRCE** We wish to solve Equation 3 sparsely. This removes small coefficients that may arise due to sensor noise as well as weak dependencies that may not contribute significantly to the VAR models. This produces a sparse, more readable graph. We also wish to solve jointly for  $\mathbf{B}$  and  $\Omega$ , as underfitting  $\mathbf{B}$  may artificially shrink values in  $\Omega$ , and overfitting may propagate errors and create an unstable precision matrix  $\Omega$ . For all of these reasons we use Multivariate Regression with Covariance Estimation (MRCE). This method was introduced by Rothman et al. [22]. They jointly fit the coefficient matrix  $\mathbf{B}$  and  $\Omega$  with lasso penalty terms. More formally:

$$(\hat{\mathbf{B}}, \hat{\Omega}) = \arg \min_{\mathbf{B}, \Omega} \{g(\mathbf{B}, \Omega) + \lambda_1 \sum_{j' \neq j} |\omega_{j',j}| + \lambda_2 \sum_{j=1}^{qp} \sum_{k=1}^q |b_{jk}|\} \quad (12)$$

where  $g$  is the negative log-likelihood function of  $(\mathbf{B}, \Omega)$ :

$$g(\mathbf{B}, \Omega) = \text{tr} \left[ \frac{1}{n} (\mathbf{Y} - \mathbf{XB})^\top (\mathbf{Y} - \mathbf{XB}) \Omega \right] - \log |\Omega| \quad (13)$$

and  $\lambda_1, \lambda_2 \geq 0$  are tuning parameters. The output is an estimate of a sparse Vector Autoregression (sVAR) coefficient matrix  $\hat{\mathbf{B}}$ , which encodes temporal dependencies between variables, and an inverse covariance matrix  $\hat{\Omega}$  which encodes the conditional dependencies between sensors. Rothman et al. propose an algorithm in which  $\hat{\mathbf{B}}$  is updated using gradient descent, and  $\hat{\Omega}$  is updated using Graphical LASSO [9]. This is done in a loop until a stopping criterion is reached.

To choose the parameters  $\lambda_1$  and  $\lambda_2$ , we adopt the following K-fold cross-validation approach recommended by Rothman et al. [22]:

$$(\hat{\lambda}_1, \hat{\lambda}_2) = \arg \min_{\lambda_1, \lambda_2} \sum_{k=1}^K \| \mathbf{Y}^{(k)} - \mathbf{X}^{(k)} \mathbf{B}_{\lambda_1, \lambda_2}^{(-k)} \|_F^2 \quad (14)$$

where  $\mathbf{B}_{\lambda_1, \lambda_2}^{(-k)}$  is the estimated coefficient matrix fit using data outside the  $k$ th fold. The authors note that  $\lambda_1$ , which penalizes regression coefficients, has a bigger impact on this objective function than  $\lambda_2$ . This is natural when  $\lambda_1$  reduces the number of coefficients, and the predictive accuracy of the VAR component.



## 4 Experiments

### 4.1 Data and Preprocessing

We use data from a wastewater treatment plant in Ireland, from the 1st of June 2025 until the 30th of June 2026. Sensor data is sampled from the aeration tank. We selected sensors installed in the middle of the tank, near the tank wall, as they are less sensitive to changes in nutrient loads in the influent. We have selected the DO sensor, Oxidation-Reduction Potential (ORP), Nitrate (NO<sub>3</sub>), Ammonia (AN), pH, Nitrous Oxide (N<sub>2</sub>O), and Nitrite (NO<sub>2</sub>). All sensors sample at a frequency of 1 minute, with the exception of N<sub>2</sub>O, which we downsample from 30 seconds to 1 minute. These sensors can indicate the two key processes that happen in the aeration tank, nitrification and denitrification.

### 4.2 Experiment Outline

Our hypothesis is that a good choice of segmentation should distinguish between different tank behaviours, and so create distinct graphs. Furthermore the graph structure within a segment type should be consistent if a segmentation rule is reliable.

We formalize this hypothesis in the following way. For each segmentation rule,  $R_{\text{mean},3}$ ,  $R_{\text{rank},5}$ ,  $R_{\text{rand},3}$ , and  $R_{\text{rand},5}$ , we use MRCE to derive two graphs: a temporal graph and an instantaneous graph. We reserve two thirds of the data to tune the sparsity parameters and one third for testing. To prevent seasonality from confounding results we select two months for fitting, then reserve the following month for testing, and repeat this until the data is covered. We use a blocked cross-validation approach since we have temporally ordered data. Fitting is done using the optimization function in equation 14. After finding the  $\lambda_1, \lambda_2$  which minimize this equation, we choose the sparsest pair within one standard deviation of the mean cross-validation error of this true minimum. We use this "1-SE" heuristic to prevent overfitting and avoid solutions based on possibly erroneous sensor data.

After tuning, we evaluate using bootstrap resampling. For regime  $\phi$ , let  $K_\phi$  denote the number of available segments. In each bootstrap replicate, we sample  $I_\phi$  segment instances with replacement, where  $|I_\phi| = 0.3K_\phi$ . The sampled instances are pooled and used to estimate one temporal graph and one instantaneous graph. This process is repeated 100 times for each regime, producing an empirical edge-weight distribution for every variable-pair, i.e. every edge.

## 5 Results

### 5.1 Segmentation Results

Table 1 shows the proportion of segments created which pass an Augmented Dickey-Fuller test for stationarity at a 5% significance level. Figure 4 shows the

Table 1: ADF stationarity test results.

| Segmentation        | Total tested | Total stationary |
|---------------------|--------------|------------------|
| $R_{\text{mean},3}$ | 1855         | 1411             |
| $R_{\text{rand},3}$ | 21           | 4                |
| $R_{\text{rank},5}$ | 2611         | 2577             |
| $R_{\text{rand},5}$ | 577          | 369              |

graphs derived from each segmentation rule. Weights for each segment type can be found in Table 2.

The most noticeable result is that some graphs end up being far less sparse than others. This is seen most dramatically in Fig. 4a where the high-o2-on and high-o2-off phases have far more temporal edges than the others. Furthermore there are always far more temporal edges (blue) than instantaneous edges (orange). This is natural considering Equation 14 explicitly minimizes prediction error. This was noted by Rothman et al. [22]: the VAR penalization term  $\lambda_1$  has a greater influence on prediction performance than the inverse error covariance penalization term  $\lambda_2$ . Table 2 both show that  $\lambda_2$  often has a large weight, indeed often the largest value available in the grid-search.

The small values of  $\lambda_1$  indicate that VAR performance is generally improved by including more variables and this is preferred over truly sparse configurations. This seems to depend on phase though, as both  $R_{\text{rank},5}$   $R_{\text{mean},3}$  produce sparser graphs in the low and anoxic/denitrification phases. A contributing factor may be that there are simply fewer processes happening with great magnitude during these phases, either because of a lack of oxygen in the tank or perhaps changes in influent that often coincide with the high and low regimes.

Another observation that may contribute to the lack of sparsity in temporal graphs is that almost all of the segments produced by  $R_{\text{rank},5}$  and  $R_{\text{mean},3}$  are stationary, with almost 99% and 76% segments passing the test respectively. Stationary segments are likely to be modelled more effectively by autoregression which may contribute to a lower regularization for this component of the MRCE method.

## 5.2 Graph Stability

To assess the level of uncertainty present in the edge-weights produced by this method we perform bootstrap resampling. Our hypothesis is that correct segmentation (in the sense that it separates truly distinct phases of operation within the aeration tank) leads to more stable graphs. A stable sparse graph should assign edge-weights predictably, i.e. without uniformity or multimodality. This leads us to this study of edge-weight distribution. With 7 sensors there are 42 possible directed edges and 21 possible undirected edges, so we will study several representative examples only. Figure 5 shows the distribution of weights for the instantaneous edge between AN and Nitrate, as calculated across all segment types. Ammonia is converted to Nitrate by nitrification, a process that is biased to occur by the presence of oxygen. In the first plot ( $R_{\text{rank},5}$ ) we see

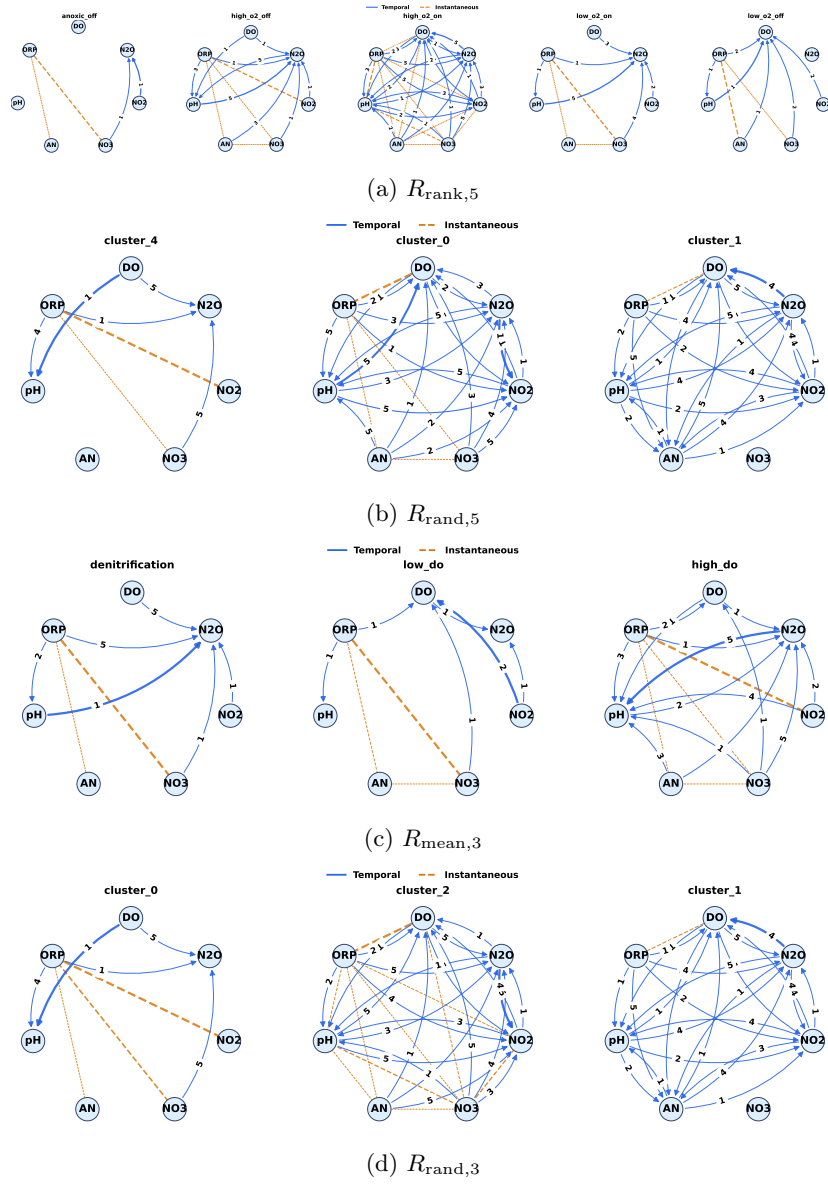


Fig. 4

that there is a higher conditional dependence between these two variables in the high-o2 phases, with a slight right-skew when aeration is on (high-o2-on). Next is the low-o2-on phase, and finally anoxic and low-o2-off are both zero. Furthermore high-o2-on/off and low-o2-on show signs of central tendency. In Fig. 5, top-right, we see a similar effect where the high-DO edge weights are clustered around  $1.4 \times 10^{-5}$ , low-DO near zero, and denitrification mostly in-between. In

| Phase       | $\lambda_1$ | $\lambda_2$ |
|-------------|-------------|-------------|
| anoxic-off  | 0.72        | 100         |
| high-o2-off | 0.14        | 100         |
| high-o2-on  | 0.0052      | 0.72        |
| low-o2-on   | 0.14        | 100         |
| low-o2-off  | 0.72        | 100         |

(a)  $R_{\text{rank},5}$ 

| Phase     | $\lambda_1$ | $\lambda_2$ |
|-----------|-------------|-------------|
| cluster-0 | 0.027       | 3.73        |
| cluster-1 | 0.027       | 3.73        |
| cluster-4 | 0.14        | 100         |
| cluster-2 | skipped     |             |
| cluster-3 | skipped     |             |

(b)  $R_{\text{rand},5}$ 

| Phase           | $\lambda_1$ | $\lambda_2$ |
|-----------------|-------------|-------------|
| denitrification | 0.14        | 100         |
| low-DO          | 0.14        | 100         |
| high-DO         | 0.027       | 100         |

(c)  $R_{\text{mean},3}$ 

| Phase     | $\lambda_1$ | $\lambda_2$ |
|-----------|-------------|-------------|
| cluster-0 | 0.14        | 100         |
| cluster-1 | 0.027       | 3.73        |
| cluster-2 | 0.027       | 0.72        |

(d)  $R_{\text{rand},3}$ 

Table 2: CV-selected MRCE penalties per phase, 5- and 3-regime segmentations.

both of the random-segmentation cases (Fig. 5, bottom) the distribution assigns incredibly high weights to one cluster while setting the others to zero.

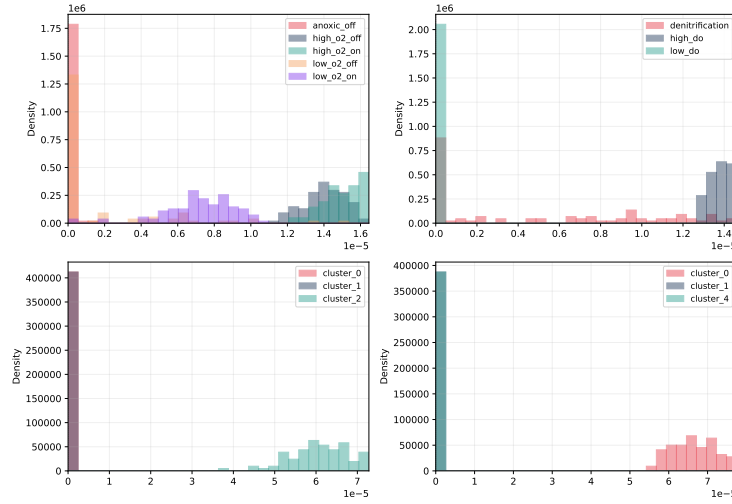
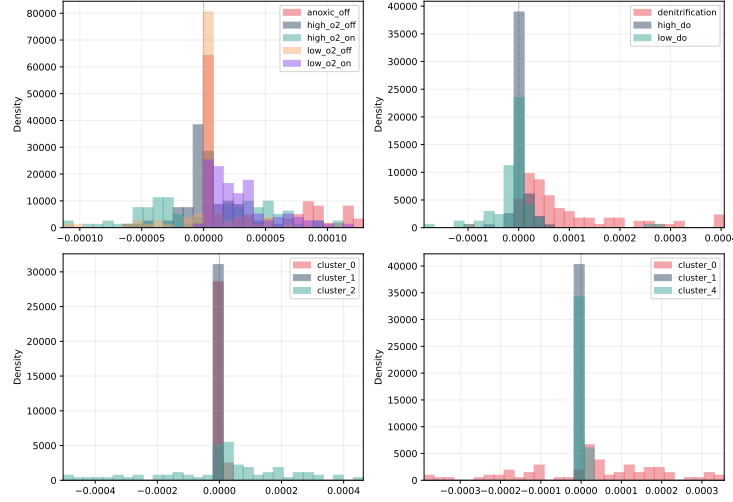


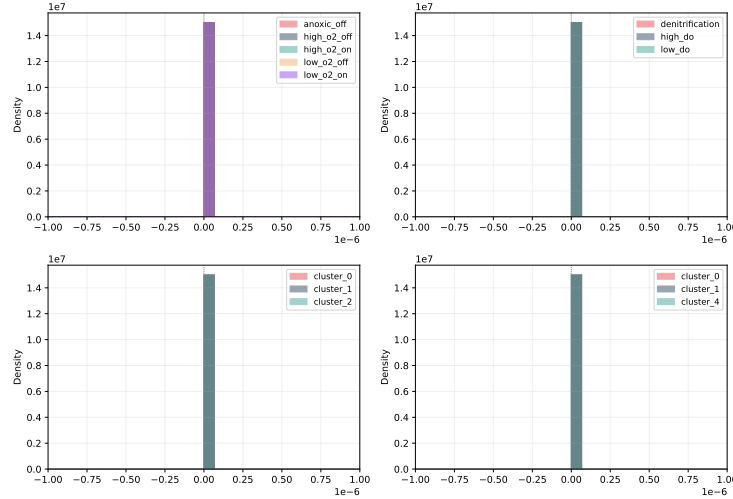
Fig. 5: Undirected edge-weight distribution of AN - Nitrate edges across all segmentation methods and segment types. Top-left:  $R_{\text{rank},5}$ ; top-right:  $R_{\text{mean},3}$ ; bottom-left:  $R_{\text{rand},5}$ ; bottom-right:  $R_{\text{rand},3}$ .

Figure 6 shows the distribution for the directed edges between Nitrate and N2O. In denitrification Nitrate is converted into Nitric Oxide (NO) and then into N2O. In the top-left plot of Fig. 6a, the largest positive weights can be seen during anoxic-off; conversely larger negative weights occur for high-o2-on. This

aligns with expected behaviour, and shows that Nitrate Granger-causes N2O. Similarly denitrification has the largest weights in the top-right plot.



(a) Nitrate  $\rightarrow$  N2O



(b) N2O  $\rightarrow$  Nitrate

Fig. 6: Directed edge-weight distributions between N2O and Nitrate

## 6 Discussion

The results broadly support our original hypothesis, that is, an effective segmentation should create graphs that are distinct between segment types and consistent within a segment type. This is seen most clearly in Fig. 6a. The random

segmentations were either all zero or had roughly uniform edge-weight distributions. Furthermore the temporal relationship only goes in the direction we expect, with  $\text{N}_2\text{O}$  being Granger-caused by Nitrate but not vice-versa. Weights are strongly positive during low oxygen phases like anoxic-off and denitrification and negative during high-o<sub>2</sub>-on and high-DO. This aligns with well established chemical processes: denitrification occurs most strongly during anoxic conditions. The negative edge-weights likely don’t come from a direct relationship in high oxygen phases, however. In high oxygen phases Nitrate is produced from Ammonia, and  $\text{N}_2\text{O}$  can be stripped into the atmosphere by aeration and mixing. This creates the illusion of the decrease of one driving the increase of another. The edge-weights in these conditions are meaningfully smaller than in anoxic conditions for both  $R_{\text{rank},5}$ , and  $R_{\text{mean},3}$ . These results are consistent with known tank behaviours and support the claim the our segmentations capture meaningful structures within the data.

The variance in sparsity is partially explained by Table 2. In  $R_{\text{rank},5}$  high-o<sub>2</sub>-on has the smallest  $\lambda_1$  and the densest temporal graph (Fig. 4a) while anoxic-off and low-o<sub>2</sub>-off receive the largest penalties with the sparsest graphs. This is also seen in  $R_{\text{mean},3}$ , where high-DO has the smallest  $\lambda_1$  and least sparse graph. High-DO in Fig. 4c contains 13 temporal edges, whilst high-o<sub>2</sub>-on and off in Fig. 4a contain 9 and 19 respectively. This discrepancy is explained by the weights of  $\lambda_1$ ; this is, we believe, due to a higher number of stationary segments in  $R_{\text{rank},5}$ . Stationary processes are better modelled by autoregressive methods which may allow for a smaller regularization term without overfitting. All of this together supports the idea that MRCE is a highly flexible method to induce graphs, being able to handle segment types with dynamic behaviour as well as segment types with fewer active processes.

The instantaneous graphs are far more sparse, with  $\lambda_2$  being extremely high in almost every configuration. The edge-weight distribution shown in Fig. 5 shows a strong right skew in high-o<sub>2</sub>-on, less so in high-o<sub>2</sub>-off, a somewhat symmetric distribution in low-o<sub>2</sub>-on, and zero in the low-o<sub>2</sub>-off and anoxic. This edge represents conditional dependence between two sensors’ residuals after VAR has been used to forecast. This has been named “instantaneous” here, and “contemporaneous” in other works [7], but in principle this is dependence that remains unexplained by VAR. This could be due to non-temporal relationships between parameters. It is far more likely that this indicates a non-linear relationship. Indeed wastewater kinetics are known to be non-linear [13], and simple VAR may not capture this. This could be improved by use of a non-linear model, such as MLP-based Granger Causality as outlined in Rosol et al. [21].

## 7 Conclusions

In this paper we proposed DO-based segmentation to create more stationary and modelable segments of wastewater sensor data. To assess these segmentations and identify key sensor relationships we used Gaussian graphical models to create temporal and instantaneous graph structures per-segment. These graphs align

with known behaviours in the aeration tank. We inferred the graphs using sparse Multivariate Regression with Covariance Estimation, which we have shown is robust to the varying dynamics of each segment type.

To quantify uncertainty we use bootstrap resampling to find the empirical edge-weight distribution for each edge. This showed that segment types were consistent and meaningfully distinct from each other. In this aspect our DO-based rules outperformed the random baselines.

Future work may involve more detailed analysis of the MRCE graphs for anomaly detection. Bayesian approaches can quantify uncertainty natively and may remove the need for bootstrap resampling [27]. Non-linear Granger causality could be integrated into the MRCE optimization algorithm.

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## References

1. Adams, R.P., MacKay, D.J.: Bayesian online changepoint detection. arXiv preprint arXiv:0710.3742 (2007)
2. Assef, F.M., Steiner, M.T.A., de Lima, E.P.: A review of clustering techniques for waste management. *Heliyon* **8**(1) (2022)
3. Ba, A., McKenna, S.A.: Water quality monitoring with online change-point detection methods. *Journal of Hydroinformatics* **17**(1), 7–19 (2015)
4. Betken, A., Micali, G., Schmidt-Hieber, J.: Ordinal pattern-based change point detection. *Test* **34**(4), 927–980 (2025)
5. Czarnota, J., Masłoń, A., Pajura, R.: Wastewater treatment plants as a source of malodorous substances hazardous to health, including a case study from Poland. *Int. J. Environmental Research and Public Health* **20**(7), 5379 (2023)
6. Daigger, G.T., Littleton, H.X.: Simultaneous biological nutrient removal: A state-of-the-art review. *Water Environment Research* **86**(3), 245–257 (2014)
7. Epskamp, S., Waldorp, L.J., Möttus, R., Borsboom, D.: The Gaussian graphical model in cross-sectional and time-series data. *Multivariate behavioral research* **53**(4), 453–480 (2018)
8. European Commission: Commission decides to refer Ireland to the Court of Justice of the European Union for failing to comply with EU rules for treating urban waste water. Press release IP/25/2957 (Dec 2025), accessed: 2026-06-03
9. Friedman, J., Hastie, T., Tibshirani, R.: Sparse inverse covariance estimation with the graphical lasso. *Biostatistics* **9**(3), 432–441 (2008)
10. Gålfalk, M., Bastviken, D.: In situ observations reveal underestimated greenhouse gas emissions from wastewater treatment with anaerobic digestion—sludge was a major source for both CH<sub>4</sub> and N<sub>2</sub>O. *Environmental Science & Technology* **59**(34), 18146–18155 (2025)
11. Gernaey, K.V., Rosén, C., Jeppsson, U.: BSM2: A model for dynamic influent data generation. Tech. Rep. CODEN:LUTEDX/(TEIE-7231)/1-73/(2005), Dept. of Industrial Electrical Engineering and Automation Lund University (2005), <https://www.iea.lth.se/publications/Reports/LTH-IEA-7231.pdf>

12. Granger, C.W.: Investigating causal relations by econometric models and cross-spectral methods. *Econometrica* pp. 424–438 (1969)
13. Haimi, H., Mulas, M., Corona, F., Vahala, R.: Data-derived soft-sensors for biological wastewater treatment plants: An overview. *Environmental modelling & software* **47**, 88–107 (2013)
14. Hansen, L.D., Rani, A., Stokholm-Bjerregaard, M.A., Stentoft, P.A., Arroyo, D.O., Durdevic, P.: Time series dataset for modeling and forecasting of n2o in wastewater treatment. *arXiv preprint arXiv:2407.05959* (2024)
15. Killick, R., Fearnhead, P., Eckley, I.A.: Optimal detection of changepoints with a linear computational cost. *JASA* **107**(500), 1590–1598 (2012)
16. Li, J., Fearnhead, P., Fryzlewicz, P., Wang, T.: Automatic change-point detection in time series via deep learning. *Journal of the Royal Statistical Society Series B: Statistical Methodology* **86**(2), 273–285 (2024)
17. Lütkepohl, H.: VAR order selection and checking the model adequacy. In: *New introduction to multiple time series analysis*, pp. 135–192. Springer (2005)
18. Newhart, K.B., Holloway, R.W., Hering, A.S., Cath, T.Y.: Data-driven performance analyses of wastewater treatment plants: A review. *Water research* **157**, 498–513 (2019)
19. Oluseyi, T., Nweke, J.: Implications of improper sewage management on public health: A case study of Kosofe Local Government Area, Lagos State. *African Journal of Housing and Sustainable Development* **1**(1) (2020)
20. Pamfil, R., Sriwattanaworachai, N., Desai, S., Pilgerstorfer, P., Georgatzis, K., Beaumont, P., Aragam, B.: Dynotears: Structure learning from time-series data. In: *Int. conf. artificial intelligence and statistics*. pp. 1595–1605. Pmlr (2020)
21. Rosoł, M., Młynczak, M., Cybulski, G.: Granger causality test with non-linear neural-network-based methods: Python package and simulation study. *Computer Methods and Programs in Biomedicine* **216**, 106669 (2022). <https://doi.org/10.1016/j.cmpb.2022.106669>
22. Rothman, A.J., Levina, E., Zhu, J.: Sparse multivariate regression with covariance estimation. *Journal of Computational and Graphical Statistics* **19**(4), 947–962 (2010)
23. Runge, J., Gerhardus, A., Varando, G., Eyring, V., Camps-Valls, G.: Causal inference for time series. *Nature Reviews Earth & Environment* **4**(7), 487–505 (2023)
24. Song, C., Ponder, D., Peng, W., Ren, Z.J.: Discrepancies in national inventories reveal a large emissions gap in the wastewater sector. *Nature Climate Change* pp. 1–9 (2026)
25. Wareham, D.G., Hall, K.J., Mavinic, D.S.: Real-time control of wastewater treatment systems using ORP. *Water science and technology* **28**(11-12), 273–282 (1993)
26. Xu, R., Song, Z., Wu, J., Wang, C., Zhou, S.: Change-point detection with deep learning: A review. *Frontiers of Engineering Management* **12**(1), 154–176 (2025)
27. Zhang, Y., Chen, J.: Alternative bayesian regularizations for factor correlations and loadings: Lasso, spike-and-slab, and horseshoe. *Structural Equation Modeling: A Multidisciplinary Journal* pp. 1–19 (2026)